1 I

1.1 (1)

The Lagrangian is defined as:

$$L = T - U$$

Since the system contains only one free particle, its potential energy is zero:

$$T = \frac{1}{2}m|\vec{V}|^2, \qquad U = 0$$

Thus, we obtain:

$$L = \frac{1}{2}m|\vec{V}|^2$$

1.2 (2)

Since the motion is confined to the XY (or xy) plane, we have $\dot{Z} = \dot{z} = 0$. Therefore:

$$\vec{V} = \vec{v} + \vec{\Omega} \times \vec{r} = (\dot{x}, \dot{y}, 0) + (-\Omega y, \Omega x, 0)$$

Then, we can obtain the Lagrangian in the rotating reference frame:

$$L = \frac{1}{2}m\left[\dot{x}^2 + \dot{y}^2 + \Omega^2(x^2 + y^2) + 2\Omega(-y\dot{x} + x\dot{y})\right]$$

1.3 (3)

Applying the Euler–Lagrange equations

$$\frac{\partial L}{\partial q} = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}$$

to x and y, respectively, we obtain:

$$\ddot{x} = \Omega^2 x + 2\Omega\dot{y},$$
$$\ddot{y} = \Omega^2 y - 2\Omega\dot{x}$$

2 II

2.1 (4)

Since the origin is the center of mass of the two-body system:

$$x_A M_A + x_B M_B = 0$$

We also know that the distance between the two particles is:

$$a = x_B - x_A$$

Using the definition of β, we can finally obtain:

$$x_A = -a\frac{M_B}{M_A + M_B} = -\beta a,$$
$$x_B = a + x_A = (1 - \beta)a$$

2.2 (5)

The centrifugal force acting on particle B is:

$$\vec{F}_{cB} = -M_B\vec{\omega} \times (\vec{\omega} \times \vec{r}_B) = M_B\omega^2 x_B\hat{x}$$

The gravitational force exerted by particle A on particle B is:

$$\vec{F}_{gB} = -\frac{GM_AM_B}{a^2}\hat{x}$$

Since particles A and B are in equilibrium in the rotating reference frame:

$$\vec{F}_{cB} + \vec{F}_{gB} = 0$$

Therefore:

$$M_B\omega^2 x_B = \frac{GM_AM_B}{a^2}$$

Substituting $x_B = (1-\beta)a$, we obtain:

$$\omega^2 = \frac{GM_A}{a^2x_B} = \frac{G(M_A+M_B)}{a^3}$$

2.3 (6)

Similarly, considering the balance between the centrifugal force and the gravitational forces acting on particle P, we obtain:

$$m\omega^2 x_L = \frac{GM_Am}{(x_L-x_A)^2} + \frac{GM_Bm}{(x_L-x_B)^2}$$

2.4 (7)

Substituting the results from (4) and (5) into the equation obtained in (6), we obtain:

$$\tilde{x} = \frac{1-\beta}{(\tilde{x}+\beta)^2} + \frac{\beta}{(\tilde{x}-(1-\beta))^2}$$

2.5 (8)

Since $x_L = x_B + \delta a$, we have:

$$\tilde{x} = 1 - \beta + \delta$$

Substituting this relation into the equation obtained in (7):

$$1-\beta+\delta = \frac{1-\beta}{(1+\delta)^2} + \frac{\beta}{\delta^2}$$

Solving for β, we obtain:

$$\beta = \frac{\delta^3(\delta^2+3\delta+3)}{\delta^4+2\delta^3+\delta^2+2\delta+1}$$

Since $\delta \ll 1$, the lowest-order approximation is:

$$\boxed{\beta \simeq 3\delta^3}$$

2.6 (9)

$$a(\tilde{x}-(1-\beta)) = a\delta = a(\frac{\beta}{3})^{1/3} \approx 1.5\times10^9\mathrm{m}$$